Oguz Akkas

ML & Data Engineer

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ML and data engineer who turns messy data into ML and LLM systems that run in production. Builds OCR and LLM extraction pipelines, data platforms, and MLOps tooling for healthcare, insurance, and software teams, and leads teams from design through delivery. Earlier published computer vision research at UW–Madison.

SKILLS

Programming Languages: Python, SQL, JavaScript/TypeScript, Scala

ML Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, Langchain, HuggingFace

DevOps & MLOps: Docker, Kubernetes, Terraform, MLFlow, GitHub Actions

Cloud Platforms: AWS, GCP, Databricks

Data Processing & Databases: Apache Spark, PostgreSQL, Redshift, Airflow, Pandas

Web Technologies: React, Node.js, Tailwind CSS, HTML/CSS

EXPERIENCE

Lead Data / ML Engineer · FlyteHealth — Apr 2024 – Present

- Developed Flyte Bot, a conversational interface using Langchain, OpenAI, Gemini, and more.
- Engineered OCR and LLM-based data extraction pipelines.
- Built and maintained a Data Lake and Data Warehouse on S3/Redshift, and automated CI/CD processes with Terraform.

Tools: Langchain, OpenAI, Gemini, Anthropic, Bedrock, S3, Redshift, PostgreSQL, AWS Batch, Terraform

Senior Tech Lead · Verana Health — Oct 2023 – Mar 2024

- Redesigned a commercial production tool's data pipeline with custom LLMs and GPT-4, reducing processing time from weeks to hours.
- Led efforts in defining, architecting, and developing a machine learning platform and mentored junior engineers.

Tools: LLMs, GPT-4, MLOps, Databricks, PySpark, Airflow, AWS SageMaker, Terraform, Docker

Senior ML / MLOps Engineer · Verana Health — Mar 2022 – Sep 2023

- Architected an ML platform where teams could experiment with, test, and deploy models.
- Built an automated data labeling pipeline with SageMaker Ground Truth, PyTorch and HuggingFace training templates, and a self-service deployment pipeline with automated checks, performance tests, and model monitoring.

Tools: AWS SageMaker, Ground Truth, Lambda, PyTorch, HuggingFace, Terraform, GitHub Actions, Airflow, Spark, Docker

Senior Machine Learning Software Engineer · Wind River — Apr 2021 – Mar 2022

- Led the design, development, and integration of the data science platform.
- Contributed to ETL and data extraction processes designed for distributed and parallel efficiency, and wrangled data to build and deploy analytical models.

Tools: Python, ETL, Cloud Analytics, Data Science, ML Platform

NLP Engineer / Data Scientist · Pinnacol Assurance — Sep 2019 – Apr 2021

- Performed EDA and applied various machine learning and deep learning models.
- Created end-to-end machine learning pipelines using Sklearn, Airflow, PySpark, and Prefect.
- Automated workflows with Docker, Kubernetes, and Google Cloud products.

Tools: Python, sklearn, pandas, Airflow, PySpark, Prefect, Docker, Kubernetes, GCP

EARLIER ROLES

Principal Scientist · CDC / NIOSH (via Synergy America) Sep 2018 – Sep 2019

Built text-based machine learning models that assign occupational injury classification codes from injury narratives, and ran epidemiological analyses of injury data.

Senior Research Scientist · Blankenheim Services Jan 2017 – Sep 2018

Developed and validated injury risk assessment models from ergonomic job analyses, and advised clients on safer work environments.

Machine Learning Specialist (contract) · Augment Jan 2016 – Jan 2017

Built TensorFlow and Keras deep learning models for converting XML to HTML, using Gensim Doc2Vec to vectorize markup.

Research Scientist · Occupational Ergonomics and Biomechanics Lab, UW–Madison May 2012 – Jan 2017

Used computer vision and machine learning to measure worker hand activity and driver distraction from video. Mentored 2 graduate and 5 undergraduate students.

OPEN PROJECTS

Clinical Document Extraction Demo oguzakkas.com/#live-demo

A live demo where Gemini reads synthetic lab reports and clinical notes and returns structured data, scored field by field against an answer key.

Medication & Adverse Drug Event Extraction (n2c2 2018) github.com/oakkas/n2c2_2018_task2_llm

Groundwork for LLM-based extraction of medications and adverse drug events from discharge summaries in the n2c2 2018 Track 2 clinical NLP dataset.

EDUCATION

Ph.D. (ABD), Industrial and Systems Engineering, University of Wisconsin–Madison 2017

M.S., Industrial and Systems Engineering, University of Wisconsin–Madison 2013

MBA, Istanbul University 2010

B.A., Management (Quantitative Methods), Istanbul University 2008

PUBLICATIONS

Radwin RG, Hu YH, **Akkas O**, Bao S, Harris-Adamson C, Lin JH, Meyers AR, Rempel D (2023). Comparison of the observer, single-frame video and computer vision hand activity levels. *Ergonomics 66(8)*. doi.org/10.1080/00140139.2022.2136407

Akkas O, Lee CH, Hu YH, Harris Adamson C, Rempel D, Radwin RG (2017). Measuring exertion time, duty cycle and hand activity level for industrial tasks using computer vision. *Ergonomics 60(12)*. doi.org/10.1080/00140139.2017.1346208

Radwin RG, Lee JD, **Akkas O** (2017). Driver movement patterns indicate distraction and engagement. *Human Factors 59(5)*. doi.org/10.1177/0018720817696496

Akkas O, Lee CH, Hu YH, Yen TY, Radwin RG (2016). Measuring elemental time and duty cycle using automated video processing. *Ergonomics 59(11)*. doi.org/10.1080/00140139.2016.1146347

Akkas O, Azari DP, Chen CH, Hu YH, Ulin SS, Armstrong TJ, Rempel D, Radwin RG (2015). A hand speed–duty cycle equation for estimating the ACGIH hand activity level rating. *Ergonomics 58(2)*. doi.org/10.1080/00140139.2014.966155